

Cross-Device Reproducibility of Physiological Directional Concordance in a Wearable-Derived Sugar-Trend Algorithm

A standalone retrospective multi-cohort study across Apple Watch, Samsung Galaxy Watch, and EasyTouch Rhythm Band real-world data

Original Article | Digital health • wearable physiology • metabolic trend inference • cross-device reproducibility

Participant identifiers are not reproduced. Device cohorts are analysed independently; participant counts are not summed as unique individuals because cross-cohort identity overlap was not formally resolved.

Scientific scope. This study evaluates whether a proprietary wearable-derived sugar-trend framework exhibits reproducible bidirectional, meal-associated, recovery, and physiological-state behavior across three distinct wearable ecosystems. It does not establish numerical blood-glucose accuracy, CGM equivalence, MARD, diagnostic performance, or interchangeability with a glucometer.

Structured Abstract

Background: Consumer wearables continuously capture physiological signals but do not directly measure interstitial glucose. A credible metabolic trend-inference framework should therefore demonstrate reproducible physiological behavior across different hardware ecosystems rather than depend on one device, one cohort, or one favorable temporal alignment.

Objective: To evaluate cross-device reproducibility of the Agatsa One sugar-trend algorithm across Apple Watch, Samsung Galaxy Watch, and EasyTouch Rhythm Band cohorts using four complementary properties: ordinary bidirectional transition symmetry, meal-associated upward excursions, post-peak recovery, and orthogonal fasting-to-postprandial reference-state ordering.

Design: Retrospective multi-cohort analysis of de-identified real-world commercial-use data. Each device cohort was analysed separately using harmonised transition definitions and conceptually aligned event windows, followed by a cross-device reproducibility synthesis. Calibration worksheets and calibration-flagged observations were excluded from orthogonal reference analyses.

Data: Apple Watch contributed 19,379 observations from 145 users and 3,401 meal logs; Galaxy Watch contributed 2,939 observations from 28 users and 837 meals; EasyTouch Rhythm Band contributed 83,090 observations from 128 users and 3,190 meals. Cohort counts are reported separately because possible cross-device participant overlap was not formally resolved.

Results: Across 15–90-minute ordinary transitions, rise:fall ratios were 1.006 for Apple Watch (5,747 rises; 5,715 falls), 0.978 for Galaxy Watch (614; 628), and 0.986 for Rhythm Band (23,539; 23,864). Primary meal-associated positive-event proportions were 61.9% (206/333), 53.7% (36/67), and 66.2% (1,395/2,108), respectively. Thirty-minute shifted analyses yielded 61.9%, 29.9%, and 63.5%, revealing high robustness in Apple Watch and Rhythm Band but substantial temporal sensitivity in the smaller Galaxy cohort. Post-peak recovery occurred in 91.3% (272/298), 63.9% (39/61), and 78.0% (1,596/2,046) of evaluable trajectories. After excluding calibration data, higher median post-meal than fasting manual/photo reference values were present in 25/26 (96.2%), 10/10 (100%), and 20/21 (95.2%) evaluable participants, with median within-participant differences of approximately +29, +42, and +26 mg/dL.

Interpretation: The most reproducible feature was near-unity rise/fall balance across all three hardware ecosystems. Meal-associated directionality, recovery, and orthogonal physiological-state ordering were also represented in every cohort, although event anchoring was more timing-sensitive in sparse Galaxy Watch data. These convergent findings support device-spanning physiological directional responsiveness of the Agatsa One framework while remaining distinct from numerical glucose validation.

Research in Context

Evidence before this study

Postprandial metabolic responses vary substantially between individuals and between meals, motivating interest in personalised metabolic phenotyping. Continuous glucose monitoring provides direct interstitial-glucose trajectories, whereas consumer wearables collect device-dependent physiological and contextual data streams without directly measuring glucose. Consequently, an inference framework derived from wearable data requires validation strategies that distinguish plausible dynamic physiology from numerical glucose measurement. Prior nutrition and CGM studies have established substantial inter-individual heterogeneity in postprandial responses, while international consensus documents define the distinct role of reference glucose data for quantitative glucose interpretation.

Added value of this study

Rather than relying on one wearable platform, this study tests whether the same proprietary inference framework repeatedly expresses the same structural properties across three distinct real-world wearable ecosystems. The analysis retains positive, negative, and stable transitions; uses meals as independently time-stamped physiological anchors; explicitly perturbs temporal alignment; examines post-peak recovery; and adds a separate non-calibration reference stream to confirm that fasting-to-postprandial state transitions were present in the underlying populations. The cross-device comparison therefore evaluates reproducibility of behavior rather than numerical equivalence to glucose.

Implications of all available evidence

A wearable-derived metabolic trend signal becomes more credible when bidirectional dynamics, event-associated directionality, temporal coherence, and recovery reproduce across different devices and datasets. The present findings support that intermediate feasibility layer. They do not remove the need for synchronized biochemical reference data before making numerical glucose-accuracy claims.

Introduction

Wearable devices have created unusually dense streams of real-world physiological and contextual data. The scientific opportunity is not limited to reproducing measurements already available from conventional devices; it also includes deriving higher-order physiological states through proprietary inference frameworks. The specific input channels, derived features, preprocessing, fusion logic, model parameters, training targets, and architecture used by Agatsa One are proprietary Agatsa intellectual property and are intentionally outside the scope of this manuscript. The evidentiary standard for an inferred state must nevertheless be carefully matched to what is actually measured.

Glucose is a particularly challenging target. Apple Watch, Samsung Galaxy Watch, and EasyTouch Rhythm Band do not, in the datasets analysed here, provide a direct biochemical measurement of blood or interstitial glucose. A proprietary algorithm may nevertheless generate a metabolic trend variable from wearable physiology. The appropriate first scientific question is therefore not whether that variable equals a glucometer or CGM concentration. It is whether the inferred signal behaves in ways that are directionally, temporally, and physiologically coherent.

Meals provide one useful external anchor because postprandial physiology commonly includes an upward glucose excursion followed by recovery, while the exact magnitude and timing vary by meal composition, baseline state, activity, individual phenotype, and measurement timing. This heterogeneity is important: a valid real-world analysis should not expect every meal to generate an identical spike, nor should it remove flat or negative events merely because they do not fit an idealised response.

A second requirement is bidirectionality. A trend algorithm that simply drifts upward, is mechanically mean-reverting, or embeds an asymmetric transformation could produce visually persuasive episodes without representing two-sided physiology. Examining all ordinary consecutive transitions provides a direct test of whether material rises and falls are both represented at scale. A near-balanced distribution is not proof of metabolic validity, but it is an important guard against simple one-directional artefact.

A third requirement is reproducibility across hardware. Device-specific sampling patterns, sensor characteristics, firmware, missingness, and user behavior can create apparent associations. Reproduction of the same broad directional architecture across a third-party Apple ecosystem, a third-party Samsung ecosystem, and an Agatsa-associated Rhythm Band therefore provides a stronger feasibility test than any single-device cohort.

The present study was designed as a standalone cross-device investigation. We analysed each wearable cohort independently using harmonised transition thresholds and conceptually aligned meal windows, then compared the resulting properties without pooling participants or assuming that cohort identities were mutually exclusive. We additionally examined a separate stream of non-calibration manual/photo reference measurements to establish whether expected fasting-to-postprandial state ordering was present in each population. These reference observations were not treated as synchronized wearable-reference pairs and were not used to calculate glucose accuracy.

We use the term physiological-anchor concordance to describe this intermediate validation framework: concordance between the direction and temporal structure of an inferred wearable signal and independently defined physiological anchors, together with balanced two-sided dynamics and recovery behavior. The framework is intentionally narrower than biochemical accuracy validation but stronger than presenting selected anecdotal trajectories.

Methods

Study design and data provenance

This was a retrospective, observational, multi-cohort analysis of de-identified records generated during ordinary commercial use of the Agatsa One platform. No experimental intervention, meal challenge, study stipend, or research-device allocation was assigned for this analysis. Customers used their own

subscriptions and/or purchased products in ordinary use. The three device cohorts were analysed as separate datasets and then compared at the cohort level.

The Apple Watch dataset covered 30 May to 3 September 2026 and contained 19,379 wearable observations from 145 users with 3,401 meal logs. The Galaxy Watch dataset covered 2 June to 4 September 2026 and contained 2,939 observations from 28 users with 837 meals. The Rhythm Band dataset covered 31 May to 4 September 2026 and contained 83,090 observations from 128 users with 3,190 meals. Stored timestamps were analysed as provided in Indian Standard Time. Median sampling interval was approximately 30 minutes in the Apple and Galaxy datasets; Rhythm Band generated substantially denser longitudinal sampling overall.

Because the same customer could theoretically have used more than one wearable, participant identities were not cross-resolved for this comparative manuscript. Accordingly, 145, 28, and 128 are reported as device-cohort user counts and are never summed or described as 301 unique participants.

Wearable-derived trend variable

For harmonised analysis, the scalar trend value at each wearable observation was defined as the midpoint of the stored lower and upper sugar-trend interval. The units are algorithmic trend units and are not represented as directly measured blood-glucose concentrations. The proprietary production algorithm itself was not reverse engineered for this analysis; the study evaluates the behavior of its stored output.

Ordinary transition analysis

Within each participant and device cohort, consecutive wearable observations separated by 15–90 minutes were identified. A material rise was defined as a change $>+3$ trend units, a material fall as <-3 units, and approximately stable as -3 to $+3$ units inclusive. The rise:fall ratio was calculated as the number of material rises divided by the number of material falls. This analysis used all eligible ordinary transitions rather than meal-selected episodes.

Meal-event analyses

Meal logs were used as external physiological anchors. The primary immediate analysis required an eligible pre-meal baseline and a post-meal observation/peak within the harmonised immediate window used in the device-specific analyses. The event excursion was calculated as post-meal peak minus baseline. A positive event had $\Delta > 0$. A broader paired-event analysis used a wider post-meal search window to accommodate real-world timing variability. Flat and negative events were retained in all distributions.

A prespecified 30-minute shifted analysis perturbed the alignment between meal time and wearable observations. This was not intended to discover an optimal lag. Its purpose was to test whether conclusions depended strongly on one exact timestamp assumption. Device-specific differences in sensitivity to this shift were treated as findings rather than normalized away.

Pre-meal and post-peak recovery analyses

Pre-meal dynamics were evaluated when at least two eligible observations were available before a meal. Because meal time is not an anchor for when a preceding decline should begin, the pre-meal fall proportion was considered descriptive and was not interpreted as sensitivity. For post-peak recovery, an identified meal-associated peak was followed forward within the prespecified recovery window. A trajectory was classified as recovering when a subsequent eligible observation declined from that peak. Recovery magnitude was peak minus subsequent value.

Orthogonal non-calibration reference analysis

Reference-sugar records were analysed separately from the wearable signal. The Calibration worksheet was completely excluded in every cohort. Records explicitly flagged as calibration references were also excluded. To reduce ambiguity about provenance, the principal orthogonal analysis was restricted to clean manual and photo-captured reference measurements. For each participant with both fasting and post-meal observations, the median fasting reference value and median post-meal reference value were calculated. The endpoint

was whether the participant-level post-meal median exceeded the fasting median, together with the within-participant difference.

These reference measurements were not assumed to be simultaneous with wearable observations and were not used to calculate MARD, error grids, numerical bias, diagnostic sensitivity/specificity, or wearable-reference agreement. Their purpose was narrower: to verify that independently recorded physiological state transitions were present in the same device cohorts in which wearable-derived meal dynamics were evaluated.

Statistical analysis

Analyses were descriptive and designed around reproducibility rather than device ranking. Counts and proportions were calculated for transition classes, meal-event direction, and recovery. Wilson 95% confidence intervals were used for principal proportions where available. Medians were used for excursion magnitudes because free-living responses were heterogeneous and non-Gaussian. Pearson correlations between meal nutrient estimates and event excursions were considered exploratory and were not treated as validation endpoints. No formal pooled participant-level model was fitted because cohort composition, sampling density, and potential participant overlap differed by device.

Cross-device reproducibility was evaluated by comparing the direction, magnitude range, and robustness of pre-specified cohort-level endpoints. In particular, we examined whether rise:fall ratios remained close to unity; whether meal-associated positive events constituted a substantial fraction of evaluable events; whether recovery occurred in a majority of evaluable post-peak trajectories; and whether non-calibration reference states showed the expected fasting-to-postprandial ordering.

Missing data, repeated observations and multiplicity

No imputation was performed for missing wearable observations, meal logs, or reference measurements. Each endpoint used only events satisfying its own evaluability criteria. Multiple meals and transitions from the same participant were retained because the objective was to characterize real-world signal behavior; therefore event-level confidence intervals should not be interpreted as fully accounting for within-participant clustering. The analysis is exploratory/feasibility-oriented and no multiplicity-adjusted hypothesis-testing framework was applied.

Results

Cohort characteristics and observation density

Characteristic	Apple Watch	Galaxy Watch	EasyTouch Rhythm Band
Users in device cohort	145	28	128
Wearable observations	19,379	2,939	83,090
Meal logs	3,401	837	3,190
Study period	30 May–3 Sep 2026	2 Jun–4 Sep 2026	31 May–4 Sep 2026
Valid 15–90 min transitions	17,156	2,524	78,475
Primary evaluable meal events	333	67	2,108
Evaluable post-peak trajectories	298	61	2,046
Clean reference participants with fasting + post-meal	26	10	21

The cohorts differed markedly in scale and sampling density. Rhythm Band contributed the largest number of wearable observations and evaluable meal events, Apple Watch contributed the largest third-party wearable cohort, and Galaxy Watch provided a smaller and sparser dataset. These differences are central to interpretation of temporal robustness and are not treated as nuisance variation.

Cross-device bidirectional transition symmetry

Across 17,156 Apple Watch transitions, 5,747 (33.5%; 95% CI 32.8–34.2%) were material rises, 5,715 (33.3%; 95% CI 32.6–34.0%) were material falls, and 5,694 (33.2%) were stable, yielding a rise:fall ratio of 1.006. Thus, material upward and downward transitions were nearly identical in frequency.

Across 2,524 Galaxy Watch transitions, 614 (24.3%; 95% CI 22.7–26.0%) were rises, 628 (24.9%; 95% CI 23.2–26.6%) were falls, and 1,282 (50.8%) were stable, yielding a rise:fall ratio of 0.978. The higher stable fraction is consistent with the sparser/smaller cohort but the material transitions remained closely balanced.

Across 78,475 Rhythm Band transitions, 23,539 (30.0%; 95% CI 29.7–30.3%) were rises, 23,864 (30.4%; 95% CI 30.1–30.7%) were falls, and 31,072 (39.6%) were stable, yielding a rise:fall ratio of 0.986. This balance persisted over a transition count more than four times larger than the Apple cohort and more than thirty times larger than the Galaxy cohort.

The cross-device result is therefore structurally consistent: all three rise:fall ratios fell between 0.978 and 1.006. The absolute percentage of stable transitions varied substantially, but the relative balance between material upward and downward movement remained close to unity.

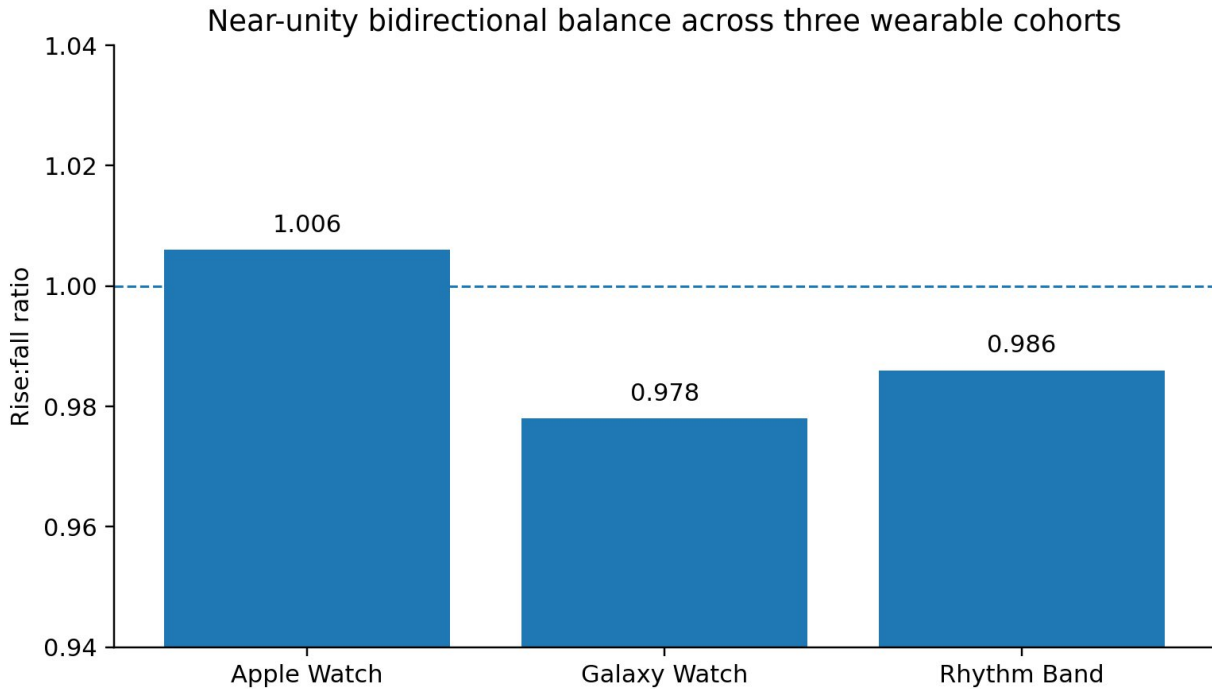


Figure 1 | Cross-device bidirectional balance. Material rise:fall ratios were close to unity in all three cohorts despite large differences in hardware, cohort size and sampling volume.

Apple Watch: meal-associated dynamics

In the Apple Watch primary raw-timestamp analysis, 333 meal events had sufficient baseline and post-meal observations; 206/333 (61.9%; 95% CI 56.5–66.9%) showed a positive post-meal excursion, with median delta +3.5 trend units. Under the 30-minute shifted alignment, 200/323 events (61.9%; 95% CI 56.5–67.0%) remained positive and the median change was +3.0. The identical positive-event proportion under the two alignments is a notable robustness result.

In the broader Apple paired-event set, 234/336 (69.6%; 95% CI 64.5–74.3%) were positive, with mean delta +7.75 and median +5.5 trend units. Large excursions were also represented: 128/336 (38.1%) reached at least +10 units and 62/336 (18.5%) reached at least +20 units. Nutrient-to-delta correlations were weak and did not support a simple linear carbohydrate-dose model, consistent with heterogeneous free-living responses.

Among 392 Apple meal events with at least two pre-meal observations, 142 (36.2%) showed a net pre-meal decline. More informative was post-peak behavior: among 298 evaluable trajectories, 272 (91.3%; 95% CI 87.5–94.0%) subsequently declined, with median peak-to-recovery reduction of 16 units; 257/298 (86.2%) declined by more than 3 units and 209/298 (70.1%) declined by at least 10 units. The resulting sequence—meal anchor, upward excursion, peak, subsequent decline—was the strongest event-level temporal pattern in the Apple cohort.

Samsung Galaxy Watch: meal-associated dynamics and temporal sensitivity

In the harmonised Galaxy primary analysis, 67 meal events were evaluable and 36/67 (53.7%; 95% CI 41.9–65.1%) showed a positive post-meal excursion; median delta was +1.5 trend units. In the broader paired-event set, 28/70 (40.0%) were positive. The full event distribution retained upward, flat and downward responses rather than selecting only visually favorable episodes.

The 30-minute shifted Galaxy analysis produced a materially different result: 23/77 (29.9%; 95% CI 20.8–40.8%) were positive. This timing dependence is important. It indicates that sparse sampling, uncertain physiological latency, meal timestamp precision, or device-specific data availability can materially affect

event classification in a small cohort. The result is therefore reported transparently rather than replaced by whichever alignment is more favorable.

Among 84 Galaxy meal events with at least two pre-meal observations, 34 (40.5%) showed a net decline. Post-peak recovery was evaluable in 61 trajectories; 39/61 (63.9%; 95% CI 51.4–74.8%) subsequently declined, with median peak-to-recovery reduction of 3.5 trend units and 31/61 (50.8%) declining by more than 3 units. Although weaker than Apple and Rhythm Band, downward recovery remained represented in a majority of evaluable trajectories.

EasyTouch Rhythm Band: meal-associated dynamics at scale

Rhythm Band provided the largest event set. In the primary analysis, 1,395/2,108 meal events (66.2%; 95% CI 64.1–68.2%) showed a positive excursion, with median +6.0 trend units. In the broader paired-event analysis, 1,580/2,177 (72.6%) were positive, with median +9.5 units. Of the broader events, 1,081 (49.7%) reached at least +10 units and 794 (36.5%) reached at least +20 units.

Under the 30-minute shifted alignment, 1,349/2,124 Rhythm events (63.5%) remained positive. The modest reduction from 66.2% to 63.5%, while retaining a clear majority of positive events, indicates greater temporal robustness than the Galaxy cohort and somewhat less exact invariance than Apple.

Among 2,117 Rhythm meal events with sufficient pre-meal observations, 940 (44.4%) showed a net decline. Post-peak recovery was evaluable in 2,046 trajectories; 1,596/2,046 (78.0%; 95% CI 76.2–79.7%) subsequently declined, with median peak-to-recovery reduction 7.75 units and 1,275/2,046 (62.3%) declining by more than 3 units. This large event count provides substantial real-world evidence that the inferred signal does not merely rise around meals but also expresses subsequent downward dynamics.

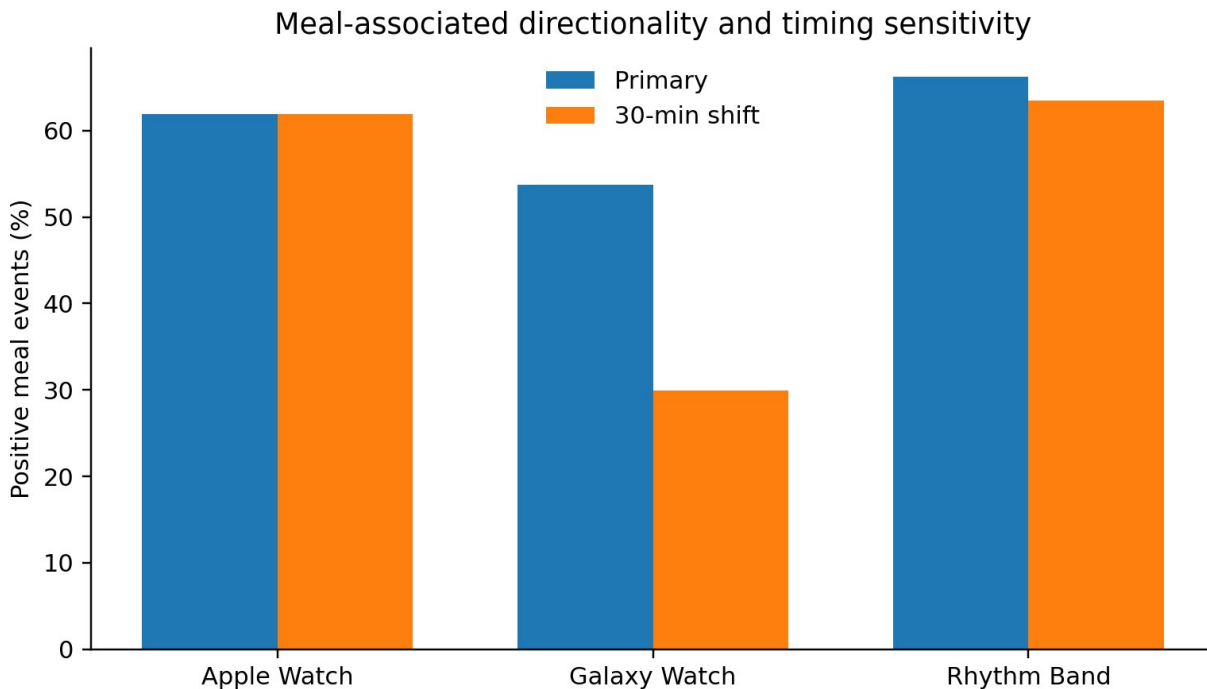


Figure 2 | Primary meal-associated directionality and 30-minute timing sensitivity. Apple Watch and Rhythm Band retained a majority of positive events after the timing perturbation; the smaller Galaxy Watch cohort was substantially more alignment-sensitive.

Post-peak recovery across devices

Recovery provides a distinct test from meal-associated rise because it asks whether the signal subsequently reverses direction after an identified post-meal peak. Recovery was observed in 272/298 Apple trajectories (91.3%), 39/61 Galaxy trajectories (63.9%), and 1,596/2,046 Rhythm trajectories (78.0%). Thus, a majority of evaluable trajectories recovered in every cohort, with the strongest proportion in Apple and the largest absolute number of recovery events in Rhythm Band.

The device differences should not be interpreted as a ranking of physiological accuracy. Recovery evaluability depends on post-peak observation availability, sampling density, event-window definitions, user behavior, and the position of the detected peak within the available time series. The cross-device conclusion is therefore qualitative and structural: downward recovery is represented after meal-associated peaks across all three ecosystems.

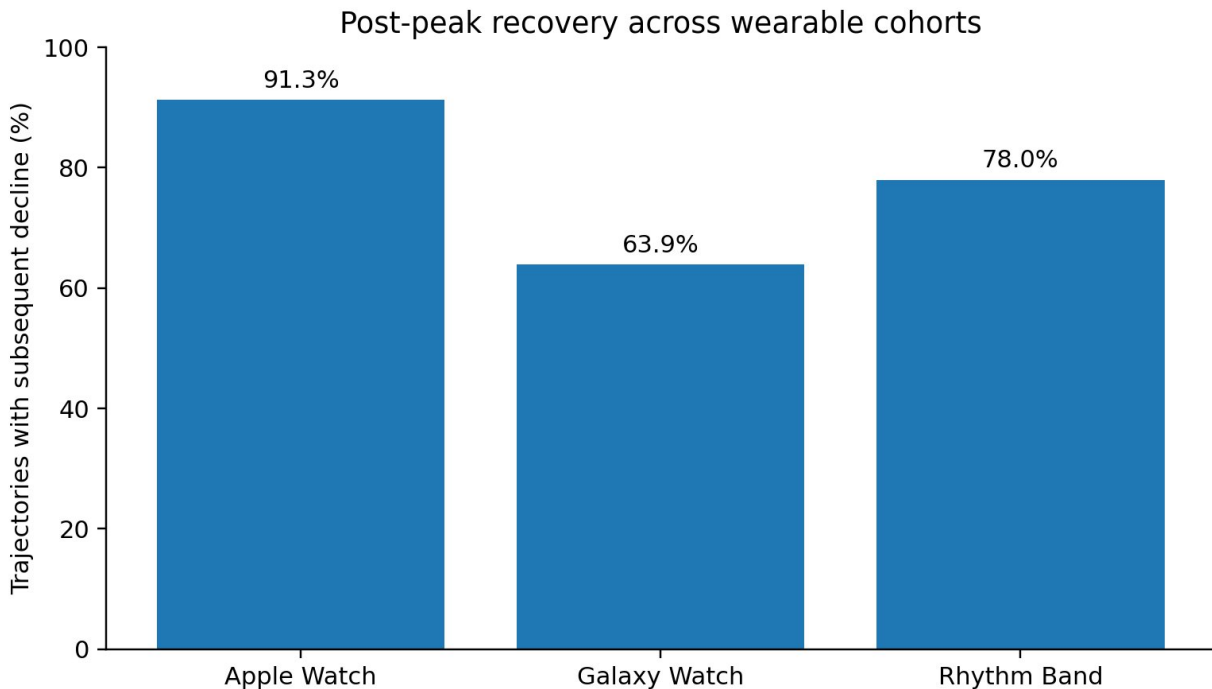


Figure 3 | Post-peak recovery across device cohorts. Subsequent downward movement was observed in a majority of evaluable trajectories in each dataset.

Orthogonal reference-state confirmation after complete calibration exclusion

The orthogonal reference analysis was deliberately separated from the wearable-event analysis. Calibration worksheets were completely excluded and calibration-flagged records were removed. The principal analysis used manual/photo-captured fasting and post-meal reference observations because these provided the clearest independent state labels without treating algorithm calibration points as validation data.

In Apple Watch users with both fasting and post-meal clean reference observations, 25/26 participants (96.2%) had a higher median post-meal than fasting reference value. The median within-participant difference was approximately +29 mg/dL. These measurements independently demonstrate that the Apple cohort contained measurable fasting-to-postprandial glucose state differences.

In the Galaxy cohort, all 10/10 evaluable participants had a higher median post-meal than fasting reference value, with median within-participant difference +42 mg/dL. The small sample size requires caution, but the direction was unanimous in the evaluable subset.

In the Rhythm cohort, 20/21 participants (95.2%) had a higher median post-meal than fasting reference value, with median within-participant difference +26 mg/dL. A stricter same-day sequence analysis identified

48 fasting-to-later-post-meal reference pairs across 12 participants; 39/48 (81.2%) increased, with median change +22 mg/dL.

These reference results are not evidence that the wearable trend numerically matched the reference values. Instead, they establish an orthogonal fact: the populations in which meal-associated wearable dynamics were observed also exhibited the expected independently measured fasting-to-postprandial state ordering. This reduces reliance on meal logs alone as the only external physiological context.

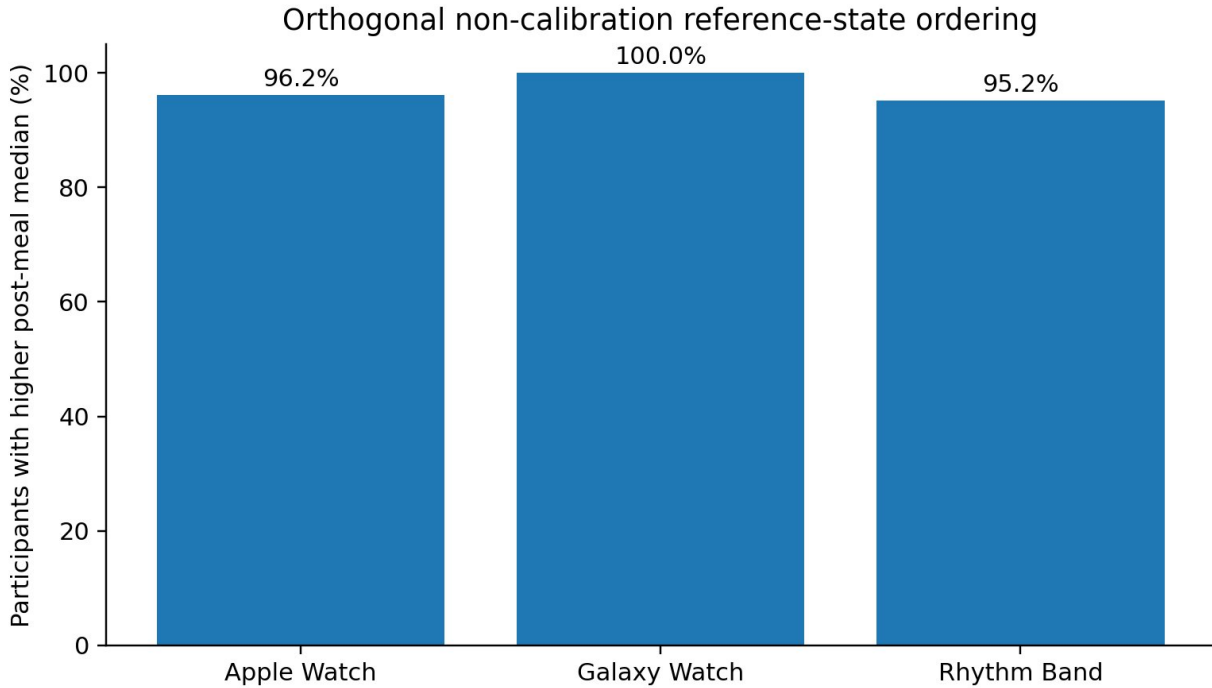


Figure 4 | Orthogonal fasting-to-postprandial reference-state ordering. After exclusion of all calibration data, nearly every evaluable participant with clean manual/photo observations had a higher median post-meal reference state.

Integrated cross-device endpoint matrix

Endpoint	Apple Watch	Galaxy Watch	Rhythm Band	Cross-device interpretation
Rise:fall ratio	1.006	0.978	0.986	Near-unity in all three
Stable transitions	33.2%	50.8%	39.6%	Device/sampling heterogeneity
Primary meal-positive	61.9%	53.7%	66.2%	Positive majority in Apple/Rhythm; modest Galaxy
30-min shifted positive	61.9%	29.9%	63.5%	Galaxy most timing-sensitive
Broader meal-positive	69.6%	40.0%	72.6%	Heterogeneous but retained full distributions
Post-peak recovery	91.3%	63.9%	78.0%	Majority in all three
Reference ordering	25/26 (96.2%)	10/10 (100%)	20/21 (95.2%)	Highly consistent physiological context
Median reference difference	+29 mg/dL	+42 mg/dL	+26 mg/dL	Independent state separation; not wearable accuracy

Convergence and heterogeneity

No single percentage is identical across devices, nor should identical values be expected from distinct free-living cohorts. The more important reproducibility pattern is that all three datasets independently satisfy the same structural tests: material rises and falls are balanced; meal-linked upward movement is represented; post-peak downward recovery is represented; and independent non-calibration reference measurements show expected fasting-to-postprandial ordering. At the same time, the Galaxy timing-shift result demonstrates a meaningful boundary condition: event-level conclusions can depend strongly on sampling and temporal alignment when the dataset is sparse.

Discussion

Principal findings

This standalone cross-device study evaluated more than 100,000 wearable observations across three device cohorts without assuming that participant identities were mutually exclusive. The most striking result was the repeated near-unity balance between material rises and falls: 1.006 for Apple Watch, 0.978 for Galaxy Watch, and 0.986 for Rhythm Band. This pattern persisted despite major differences in hardware, cohort size, stable-transition fraction, and event density.

Meal anchoring provided a second layer of evidence. Positive immediate post-meal excursions occurred in 61.9% of Apple, 53.7% of Galaxy, and 66.2% of Rhythm events. Broader event windows increased the positive proportion in Apple and Rhythm but not Galaxy. A prespecified 30-minute timing perturbation left Apple unchanged and Rhythm broadly similar, while materially reducing the Galaxy positive-event proportion. Rather than weakening the cross-device study, this contrast defines an important property of real-world wearable inference: temporal robustness itself is device- and data-density-dependent and should be reported rather than optimized post hoc.

Post-peak recovery provided a third, directionally independent test. A majority of evaluable trajectories declined after the identified meal-associated peak in every cohort, ranging from 63.9% in Galaxy to 91.3% in Apple, with 1,596 recovery trajectories in Rhythm Band. Finally, an independent non-calibration reference stream showed higher post-meal than fasting states in 96.2%, 100%, and 95.2% of evaluable participants in the Apple, Galaxy, and Rhythm cohorts, respectively. The reference stream therefore confirms the presence of genuine physiological state separation without being misrepresented as synchronized wearable validation.

Why the near-unity rise:fall ratio matters

A derived trend signal can look plausible for reasons unrelated to physiology. Smoothing, bounded transformations, baseline normalization, drift correction, and display rules can all shape apparent trajectories. The ordinary-transition analysis was designed as a simple falsification test against one particularly problematic behavior: systematic one-directional movement. Across all eligible transitions, upward and downward material changes occurred at nearly equal frequencies in all three datasets. The ratios were not merely on the same side of one; they clustered tightly around unity despite transition counts ranging from 2,524 to 78,475.

Near-unity symmetry is not sufficient to prove validity. A random process could also be symmetric. Its value arises in combination with the event analyses: the same signal that is globally balanced shows non-random temporal organization around meal anchors and subsequent recovery. In other words, the global distribution argues against monotonic bias, while event anchoring asks whether the balanced signal becomes directionally structured around an external physiological context.

Meal-associated dynamics should not be reduced to a spike-detection percentage

The observed meal-positive proportions should not be interpreted as sensitivity for postprandial glucose excursions. Meals differ in carbohydrate quantity and quality, mixed-macronutrient composition, timing, baseline glucose, preceding activity, sleep, stress, and individual physiology. User-entered meal timestamps also contain error, and sparse wearable sampling can miss the true peak. A free-living meal dataset should therefore contain positive, flat, and negative responses.

The value of the meal analysis lies in distributional enrichment and temporal coherence, not a requirement that every meal produce a positive event. Apple and Rhythm showed a majority of positive events under both primary and shifted alignments. Galaxy demonstrated that this conclusion can be fragile in a smaller, sparse cohort. Reporting that fragility is scientifically preferable to selecting an alignment that maximizes the desired result.

Recovery is a complementary physiological test

A system engineered to show meal-associated rises could still fail to reproduce subsequent downward behavior. Post-peak recovery therefore tests a different part of the trajectory. Recovery in 91.3% of Apple, 63.9% of Galaxy, and 78.0% of Rhythm trajectories shows that the inferred signal commonly reverses after an event-associated peak. Together with ordinary transition symmetry, this argues for a dynamic rather than unidirectional representation.

Role of the independent reference stream

The reference measurements require especially careful interpretation. They are useful because they are independent of the wearable-derived event signal and because calibration data were explicitly removed. Their strongest result is participant-level state ordering: nearly all evaluable participants had higher post-meal than fasting median reference values. This establishes that the study populations contained the expected physiological contrast.

However, the reference observations were not synchronized densely enough across all three cohorts to support a conventional accuracy analysis. It would therefore be inappropriate to calculate pooled MARD, Clarke/Parkes error grids, or numerical agreement from these data. The present paper uses the reference stream only as orthogonal physiological context. This distinction preserves the evidentiary value of the measurements without overstating what they can establish.

Cross-device reproducibility as an innovation

The conceptual innovation of the present study is the use of physiological-anchor concordance as a device-spanning feasibility framework. Four complementary questions are asked. First, does the inferred signal move in both directions at ordinary times? Second, does an independently time-stamped physiological event enrich for the expected direction? Third, does the signal subsequently recover in the opposite direction? Fourth, does a separate reference stream confirm that the population actually exhibits the physiological state contrast being used as context?

The framework is intentionally positioned between anecdotal visualization and biochemical accuracy validation. It can identify whether a proprietary trend signal has coherent dynamic behavior before a numerical glucose claim is considered. Its cross-device application is particularly informative because the same questions can be asked even when raw sensor stacks and sampling properties differ.

Hardware heterogeneity strengthens and constrains interpretation

Apple Watch and Samsung Galaxy Watch are third-party consumer wearable ecosystems, whereas EasyTouch Rhythm Band is an Agatsa-associated wearable platform. Reproduction across these sources reduces the likelihood that the overall pattern is unique to one hardware implementation. At the same time, the devices are not interchangeable. Sensor availability, sampling cadence, firmware behavior, data completeness, and user populations may differ. Cross-device reproducibility should therefore be understood as replication of structural behavior, not proof that all devices generate identical trajectories.

The Galaxy timing result is a useful example. A 30-minute shift changed the positive-event proportion from 53.7% to 29.9%, whereas Apple remained at 61.9% and Rhythm changed from 66.2% to 63.5%. This difference suggests that timing uncertainty can dominate event classification in sparse data. Any deployment or future validation should therefore pre-specify device-aware alignment rather than infer one universal lag from pooled data.

Nutrient correlations and physiological heterogeneity

Exploratory nutrient analyses in the device-specific datasets did not reveal a simple linear relationship between estimated carbohydrate load and wearable excursion magnitude. This should not be surprising in free-living data. Postprandial physiology reflects meal composition, gastric emptying, baseline state, insulin sensitivity, activity, circadian timing, and measurement latency. AI-derived nutrient estimates and user-entered meals also add measurement error. The absence of a simple dose-response therefore argues

against reducing the algorithm to a carbohydrate calculator, but it cannot by itself establish physiological specificity.

Strengths

- Three distinct wearable ecosystems analysed under a common conceptual framework.
- More than 100,000 wearable observations and nearly 100,000 eligible ordinary transitions across device cohorts, while avoiding an unsupported claim of 301 unique participants.
- Full inclusion of rising, stable, falling, positive, flat, and negative events rather than selecting only favorable trajectories.
- Explicit 30-minute temporal perturbation that reveals both robustness and fragility instead of optimizing lag post hoc.
- Post-peak recovery as an independent directional endpoint rather than relying only on meal-associated rises.
- Orthogonal non-calibration reference measurements with complete exclusion of the Calibration worksheet and calibration-flagged records.
- Clear separation between physiological directional concordance and numerical glucose accuracy.

Limitations

- No synchronized independent biochemical glucose series was available across the complete cohorts; numerical glucose accuracy cannot be inferred.
- Cohorts differed substantially in size, sampling density, event availability, and potentially participant characteristics.
- Participant identities were not cross-resolved across device datasets; cohort user counts cannot be summed as unique individuals.
- Repeated events are nested within users, while the present feasibility analysis is primarily descriptive and does not use a hierarchical mixed-effects model.
- Meal timestamps and meal composition were generated in ordinary use and may contain logging error; nutrient estimates are not laboratory measurements.
- The Galaxy cohort was small and showed substantial timing sensitivity, limiting precision of its event-level estimates.
- Reference measurements were useful for physiological-state ordering but were not contemporaneously paired with the complete wearable datasets.
- The exact production algorithm build, app version, device firmware, and server-side processing version should be frozen and reported at final database lock for formal submission.
- Because the algorithm is proprietary, independent external replication would further strengthen generalizability.

Clinical and regulatory interpretation

The appropriate interpretation is that the Agatsa One proprietary framework generated a dynamic wearable-derived trend signal with reproducible bidirectional, meal-associated, recovery, and orthogonal physiological-state properties across three device cohorts. The study does not demonstrate that Apple Watch, Galaxy Watch, Rhythm Band, or Agatsa One directly measures blood glucose. It does not support a numerical glucose concentration claim, diagnostic threshold claim, hypoglycaemia/hyperglycaemia diagnosis, or interchangeability with CGM or glucometer measurements.

The distinction is important for both scientific communication and product claims. Physiological directional concordance can be valuable as an exploratory metabolic trend feature while remaining categorically

different from a regulated glucose measurement. The present evidence should therefore be presented as cross-device feasibility and reproducibility of inferred trend behavior.

Conclusion

Across Apple Watch, Samsung Galaxy Watch, and EasyTouch Rhythm Band real-world cohorts, the Agatsa One sugar-trend algorithm repeatedly demonstrated balanced two-sided dynamics, with rise:fall ratios of 1.006, 0.978, and 0.986. Meal-associated upward excursions were represented in each cohort; post-peak recovery occurred in a majority of evaluable trajectories; and independent non-calibration reference measurements showed the expected higher post-meal than fasting state in 25/26, 10/10, and 20/21 evaluable participants. The convergence of these findings across substantially different hardware and sampling volumes supports physiological directional concordance as a reproducible property of the wearable-derived trend framework. The findings remain intentionally distinct from numerical blood-glucose accuracy or biochemical equivalence.

Declarations

Ethics approval and consent to participate

This manuscript reports a retrospective secondary analysis of de-identified real-world records generated during ordinary commercial use. No experimental intervention was assigned for this analysis. Before journal submission, the authors should obtain and report the applicable institutional ethics/IRB determination, waiver or exemption and reference number, as required by the target journal and jurisdiction. No ethics approval number is fabricated in this manuscript.

Consent for publication

No directly identifying participant information is included and no identifiable individual material is presented.

Data availability

The analyses are based on de-identified event-level wearable observations, time-stamped meal logs, and limited non-calibration reference-sugar streams. Calibration worksheets and calibration-flagged observations were excluded from orthogonal reference analyses. De-identified analytical data may be made available subject to privacy, commercial, contractual, and regulatory constraints.

Code and algorithm availability

The Agatsa One sugar-trend algorithm is proprietary. Analytical definitions used to derive the descriptive endpoints are reported in this manuscript. Exact production algorithm version/hash and analysis-script version should be frozen at database lock and reported in the final journal submission.

Funding

No external or internal study funding was provided for these retrospective analyses. Agatsa Software Pvt. Ltd. did not pay participants, provide study stipends, subsidize participation, or purchase devices for research use. The records arose from ordinary commercial use by customers who purchased subscriptions and/or products with their own funds. Agatsa developed and operates the commercial platform and proprietary algorithm but did not fund participant participation in these studies.

Competing interests

Agatsa One, EasyTouch Rhythm Band, and the sugar-trend algorithm are commercial/proprietary technologies of Agatsa Software Pvt. Ltd. Authors who are founders, directors, employees, shareholders, consultants, or inventors associated with Agatsa should disclose those relationships explicitly in the journal submission. Apple Inc. and Samsung Electronics are not represented as sponsors of this study; no sponsorship, endorsement, or research collaboration is inferred from use of data generated through their wearable ecosystems.

Intellectual property

The manuscript describes a proprietary wearable-derived sugar-trend algorithm and a physiological-anchor concordance framework. Publication does not determine patentability, ownership, freedom to operate, regulatory classification, or market authorization.

Author contributions

Final authorship and CRediT roles must reflect actual contributions and should be completed by author initials before submission. No funding-acquisition role is assigned merely because Agatsa operates the commercial platform.

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Supplementary Appendix 1 | Harmonised analytical definitions

Construct	Definition used in this study
Stored trend value	Midpoint of stored lower–upper sugar-trend interval.
Eligible ordinary transition	Consecutive within-user observations separated by 15–90 minutes.
Material rise	Change $>+3$ trend units.
Material fall	Change <-3 trend units.
Stable	Change from -3 to $+3$ trend units inclusive.
Primary meal event	Eligible pre-meal baseline with subsequent post-meal observation/peak in the harmonised immediate event window.
Timing sensitivity	Prespecified 30-minute perturbation of meal/wearable alignment; not an optimized lag.
Post-peak recovery	Eligible later observation lower than the identified meal-associated peak.
Orthogonal reference analysis	Calibration worksheet and calibration-flagged records excluded; clean manual/photo fasting and post-meal values used for state ordering only.
Clinical-accuracy endpoints	MARD, error grids, numerical glucose equivalence, and diagnostic performance not inferred.

Supplementary Appendix 2 | Device-specific numerical summary

Metric	Apple Watch	Galaxy Watch	Rhythm Band
Observations	19,379	2,939	83,090
Transitions	17,156	2,524	78,475
Rises	5,747	614	23,539
Falls	5,715	628	23,864
Stable	5,694	1,282	31,072
Rise:fall ratio	1.006	0.978	0.986
Primary meal n	333	67	2,108

Primary positive	206 (61.9%)	36 (53.7%)	1,395 (66.2%)
30-min shifted positive	200/323 (61.9%)	23/77 (29.9%)	1,349/2,124 (63.5%)
Broader positive	234/336 (69.6%)	28/70 (40.0%)	1,580/2,177 (72.6%)
Recovery	272/298 (91.3%)	39/61 (63.9%)	1,596/2,046 (78.0%)
Reference ordering	25/26 (96.2%)	10/10 (100%)	20/21 (95.2%)
Median reference difference	+29 mg/dL	+42 mg/dL	+26 mg/dL

Supplementary Appendix 3 | Submission-completion checklist

- ☐ Insert final author names, degrees, affiliations, and corresponding-author details.
- ☐ Complete CRediT contribution statement using author initials.
- ☐ Insert institutional ethics/IRB determination, waiver/exemption status, and reference number if required.
- ☐ Freeze and report Agatsa One production algorithm version/hash, analysis script version, app build, and relevant device/firmware versions.
- ☐ Confirm journal-specific requirements for data/code availability and proprietary algorithm disclosure.
- ☐ Confirm all commercial relationships, shareholdings, employment, directorships, and inventorship interests for every author.
- ☐ Confirm that device cohort identity overlap has either been resolved or remains explicitly undisclosed/unknown as stated.
- ☐ Perform journal-specific word-count, reference-style, figure-resolution, and reporting-guideline checks immediately before submission.